

FSOC Tracker: User Manual

AI-based virtual camera tracking system for coarse alignment of mobile Free Space Optical Communication (FSOC) terminals. Smart India Hackathon problem statement SIH26169, Department of Space / ISRO Space Applications Centre.

1. What the application does

The application simulates the coarse alignment stage of an FSOC link entirely in software.

- It draws a scene (the "screen", at least 2000 x 2000 pixels) with a moving optical beacon.
- A virtual pan-tilt camera looks at a 640 x 480 pixel window of that scene (4 x 3 degrees) and can turn at a limited speed, like a real motorised mount.
- Disturbances can be added: fog, haze, rain, low light, atmospheric turbulence, platform motion, camera vibration, and sensor noise.
- The tracker detects the beacon, measures its centre to a fraction of a pixel, predicts where it is going, and steers the camera window to keep it centred.
- Every run writes a per-frame log, a summary and an automatic performance report into one folder named `FSOC_<sim|video>_<name>_seed<N>_<date-time>` (section 7).
- For Benchmark 2, an `.mp4` video can be opened in place of the simulated scene.

2. Installation

2.1 Standalone executable (no Python needed)

Archives are built for every desktop platform in use:

Platform	Archive	Run
Windows 10 or 11, 64-bit	<code>FSOC-Tracker-windows-x64.zip</code>	<code>FSOC-Tracker.exe</code> (window); <code>FSOC-Tracker-cli.exe</code> for the command line
Linux, 64-bit (Ubuntu 22.04 or newer, Debian 12, BOSS, RHEL 9, Fedora)	<code>FSOC-Tracker-linux-x64.tar.gz</code>	<code>./FSOC-Tracker</code>
macOS 13 or newer, Intel	<code>FSOC-Tracker-macos-intel.zip</code>	<code>FSOC-Tracker</code>
macOS 13 or newer, Apple silicon	<code>FSOC-Tracker-macos-arm64.zip</code>	<code>FSOC-Tracker</code>

A demonstration video (`FSOC-Tracker-demo.mp4` , about four minutes, composed from the engine's own frames) and a 10 to 15 minute live demo script (`docs/DEMO_SCRIPT.md`) accompany the builds.

1. Download the archive for your platform from the downloads page.
2. Extract it anywhere the user can write to (Desktop, Documents, home folder). Do not run it from inside the archive.

3. Run the executable named above. The first start takes a few seconds.

Platform notes:

- Windows: SmartScreen may show "Windows protected your PC" because the executable is not code-signed. Choose More info, then Run anyway. No administrator rights are needed.
- Linux: the archive carries Qt and all Python libraries. The system needs the usual X11 or Wayland client libraries, present on every desktop install. On a minimal server image install `libxcb-cursor0 libxkbcommon-x11-0 libegl1` (Debian and Ubuntu names). If the file is not executable after extraction, run `chmod +x FSOC-Tracker`.
- macOS: Gatekeeper may say the application cannot be verified. Open System Settings, Privacy and Security, and choose Open Anyway, or run `xattr -dr com.apple.quarantine FSOC-Tracker` on the extracted folder once.

Results are written to a `results` folder next to the executable. Every archive is produced by the same build workflow, which also runs the test suite and a smoke test of the packaged executable on that platform before publishing it.

2.3 Web application

The same engine is also served as a web application, so no installation at all is needed on any platform: the project site opens it directly in a current browser (Chrome, Edge, Firefox or Safari, on Windows, Linux, macOS, or a tablet). Pick a scenario or set the parameters, press Start, and watch the scene, the camera window and the live specification tiles. Benchmark 2 videos can be uploaded from the page. The report, per-frame CSV and summary are downloadable when the run ends. One run is executed at a time per server; a second visitor sees the run in progress and can watch it. The Download desktop app button at the top right of the page recommends the archive for the visitor's computer and lists the others.

2.2 From source

Requires Python 3.11 or 3.12.

```
git clone <repository>
cd SIH169
python -m venv .venv
.venv/bin/pip install -e ".[dev]"          # Windows: .venv\Scripts\pip install -e ".[dev]"
.venv/bin/fsoc-tracker-gui                # start the desktop application
```

Optional, to retrain the AI detector: `pip install -e ".[train]"` then `python training/train_heatmap.py`.

3. The main window

```
+-----+-----+-----+
--+
```


error since acquisition (10 px or less), mean centroiding error, lock retention (target loss under 5 percent), and processing FPS (20 or more). In video mode the two error tiles read "n/a" because the video carries no ground truth.

Screen view (left picture)

The whole scene, downscaled, with a 2 degree grid. Cyan rectangle: the camera window and its centre. Orange circle: the true beacon (simulator only). Green cross: the tracker's estimate. Grey circles: other targets. Faint trails: where the beacon and the window have been. The legend is printed along the bottom edge.

Camera view (right picture)

What the camera window sees, contrast-stretched for display. Cyan cross: boresight. Dashed ring: the capture radius, green when locked. Line from the centre: the pointing error vector. Green box: this frame's detection. Orange dot: the prediction, with a fainter ring for the estimator's uncertainty. Green trail: recent estimates. A 1 degree scale bar sits bottom right. The banner shows the state, tracking error in pixels and arcseconds, confidence and which detector produced the measurement; the bottom line shows pan, tilt, the commanded rates, saturation and processing time.

Plots

- Errors: tracking error (window centre to beacon, cyan) and centroiding error (measured centre to true centre, orange). Dashed line: the 10 pixel specification.
- Gimbal command: pan and tilt rate commands with the rate limit dashed.
- Processing time per frame with the 20 FPS budget (50 ms) dashed.
- Tracker state: SEARCH, VERIFY, TRACK, COAST, REACQUIRE.

Telemetry column

Every internal quantity for the current frame: state, detector, candidates, confidence, SNR, pose, commands, detection, estimate, velocity, uncertainty, motion model probabilities, truth, errors and timing. Before a run it shows a short how-to.

4. Parameters

Every row of the problem statement parameter table has a control. Values outside the suggested range are allowed where the table says "user-defined".

Run

- Name, Seed: the seed makes a run exactly repeatable.
- Duration (s): length of a simulator run. Ignored for video input (the whole video runs).
- Extra targets: number of additional beacons (decoys) added with random paths.

Screen (rows 1 to 2)

- Width, Height: scene size in pixels. Default 2000 x 2000.
- Colour camera: renders a three-channel colour frame (PS row 2, optional); the tracker always works on luminance, so results are identical.
- Background: starfield, terrain, gradient or flat. Sky level and star density.

Camera (rows 3 to 6, 13 to 15)

- Width, Height: camera window in pixels. Default 640 x 480.
- FOV width, height: degrees. Default 4 x 3. Degrees per pixel is derived from these.
- Update rate: frames per second. Default 30.
- Max pan, Max tilt: degrees per second. Default 5.
- Max accel, Command latency: gimbal realism.
- Hard mode: the tracker sees only the pixels inside the window and must search.

Designated target (rows 7 to 12)

- Shape: square (default), circle, gaussian. Size in pixels (default 10). Peak intensity.
- Motion: line, circular, figure8, random, spiral, sinusoidal, waypoints, static. For waypoints, the Waypoints field takes screen-pixel points as `x,y; x,y; ...`; the beacon follows them at Speed and loops (the PS row 12 user-defined path).
- Speed, Radius, Period, Heading: path parameters. Start: random or centre.
- Blink: optional intensity modulation in Hz. 0 is steady.

Disturbances (rows 21 to 25)

- Salt and pepper fraction (0.10 = 10 percent), Gaussian sigma (up to 20), Poisson.
- Camera jitter: pixels per frame, up to 20.
- Atmosphere preset: clear, haze, fog, rain, lowlight. Selecting a preset fills contrast, brightness, blur and turbulence; each can then be edited.
- Platform motion: none, linear, circular, random, spiral, figure8; speed in pixels per frame (up to 20) and period.

Tracker

- Detector: hybrid (classical first, AI fills gaps), classical, cnn.
- Use picture-shift estimate: phase correlation as a vibration hint.
- Controller gains, deadband, capture radius, estimator lag, minimum confidence to acquire.
- Faint path: when nothing reaches the acquisition confidence, weak detections (threshold `faint_threshold_k`, default 3 sigma) are linked across frames and a motion-consistent chain with mean SNR above `faint_snr_min` (default 3.5) is promoted. Used automatically for dim beacons in low light; a static dim beacon is not covered by this path.
- Hard mode (Camera section) restricts the tracker to the window; SEARCH then flies an expanding square spiral of window-sized cells at the rate limit. A full sweep of a 2000 px screen at 5 deg/s takes about 12 s, so acquisition in hard mode is 3 to 12 s depending on where the beacon is.

5. Running Benchmark 2 (video input)

1. Click "Open video (Benchmark 2)" and choose the `.mp4` file. The application reads the file's real facts (displayed size with any rotation tag applied, average frame rate with a variable-rate warning, exact frame count, length) and calibrates the settings to them: the screen becomes the video's size and the update rate its frame rate. Target and disturbance settings are locked, because the video already contains them.
2. Set the camera window size, FOV and rate limits if the graders specify them; these are not in the file. Degree readouts depend on the FOV you set.
3. Click Start. The video frames are used as the scene; nothing is drawn by the simulator.
4. When the run ends, `frames.csv` contains the measured beacon centre for every frame (`det_x`, `det_y`) and `report.pdf` contains acquisition time, re-acquisition time, lock retention rate and FPS. Tracking and centroiding error against truth are not available because the video carries no ground truth.

From the command line: `fsoc-tracker video path/to/file.mp4`

6. Scenario files and the command line

A scenario is a YAML file with the same fields as the parameter panel. Example:

```
name: fog_circular
seed: 6
duration_s: 30
targets:
  - {shape: square, size_px: 10, motion: circular, radius_px: 400, period_s: 15}
disturbance:
  atmosphere: fog
  gaussian_sigma: 8
```

Commands:

```
fsoc-tracker run --scenario configs/scenarios/fog_circular.yaml [--seed 3] [--duration 20]
fsoc-tracker video path/to/file.mp4
fsoc-tracker batch --scenario configs/scenarios/*.yaml --seeds 0-49
fsoc-tracker gui
```

`batch` runs every scenario over every seed and writes `envelope.md`, a table of the worst and mean results per scenario.

Evaluator-supplied files

Benchmark Performance 1 gives you scenario parameters. Copy

`configs/scenarios/TEMPLATE_evaluator.yaml`, which lists every field with the PS row it implements, fill in the given values, save it under `configs/scenarios/`, and it appears in the

Scenario picker of both applications; or run it with `fsoc-tracker run -s <file>`. Fields the evaluators do not specify stay at the PS defaults.

Benchmark Performance 2 gives you `.mp4` files. No scenario file is needed: open the file in the desktop application, drop it on the web app, or run `fsoc-tracker video <file>`. The per-frame CSV then carries `det_x`, `det_y` (the measured centroids) for comparison with the evaluators' predefined values, and the report carries acquisition, re-acquisition, lock retention and FPS.

7. Output files

Every run gets one label and one folder named after it:

<code>results/FSOC_sim_<scenario>_seed<N>_<YYYYMMDD-HHMMSS>/</code>	simulated run
<code>results/FSOC_video_<file name>_<YYYYMMDD-HHMMSS>/</code>	Benchmark 2 run on a video

for example `results/FSOC_sim_clear_line_seed7_20260922-101530/`. The four files inside carry the same label as a prefix, so a report copied out of its folder still says which scenario, seed and time it belongs to:

File	Contents
<code><label>_frames.csv</code>	One row per frame: time, state, detection, estimate, camera pose, commands, truth, errors, processing time.
<code><label>_summary.json</code>	All metrics with their definitions and pass/fail against the specification.
<code><label>_report.pdf</code>	Three pages: specification check and metric table; time series; paths, histograms and model probabilities.
<code><label>_scenario.yaml</code>	The exact parameters used, so the run can be repeated.

With `--out <folder>` on the command line the folder is yours; the files inside are still labelled. Batch runs write `results/batch/<time>/<scenario>_seed<N>/` with labelled files and an `envelope.md` table. The web app names its downloads the same way.

8. Metric definitions

Metric	Definition
Acquisition time	First frame in TRACK with the tracked beacon within the capture radius (30 px) of the window centre. Spec: 2 s or less.
Tracking error	Distance from the true beacon to the window centre, over frames after acquisition. Spec: 10 px or less (mean).
Tracking error, vibration removed	The same with the per-frame camera vibration subtracted from the truth: the part a rate-limited gimbal can physically follow.
Centroiding error	Distance from the measured beacon centre to the true centre.

Metric	Definition
Tracked	Percentage of frames after acquisition in which the tracker held the beacon (state TRACK), whether or not the camera had it centred. A high tracked rate with a low lock rate means the camera, not the tracker, could not keep up.
Lock retention	Percentage of frames after acquisition in TRACK with the beacon inside the capture radius. Target loss is 100 minus this. Spec: loss under 5 percent.
Re-acquisition time	Time from losing lock to regaining it. Spec: 1 s or less.
FPS	1 divided by per-frame processing time, averaged. Spec: 20 or more.

9. Troubleshooting

- The window does not start on Linux: install `libxcb-cursor0` (Qt 6 requirement).
- Low frame rate: reduce the screen size, disable Poisson noise, or uncheck Real-time pacing to see the true processing speed.
- "Cannot open video": the file must be readable by OpenCV (H.264 `.mp4` is safest).
- The AI detector is not used: `models/beacon_heatmap.onnx` is missing; the classical detector runs alone. Retrain with `python training/train_heatmap.py`.